Smart Feedback & Suggestion Portal

Final Project Completion & Progress Report

PROJECT STATUS

1. Architecture & Backend (Phases 1 - 4)

- Standard modular folder structure and PostgreSQL database schema with integrity constraints.
- FastAPI REST backend with Pydantic v2 schemas, signed JWT authentication, and student/admin role guards.
- Interactive Swagger UI documentation (/api/docs) and CORS support for local and deployed clients.

2. Natural Language Processing Pipeline (Phases 5 - 9)

- Preprocessing: Tokenization, stopword cleaning with negation preservation, and WordNet lemmatization.
- Sentiment Analysis: VADER when installed, with a lightweight offline fallback.
- Classification & Urgency: Rule-based category mapping across academic, facilities, services, library, and placement categories.
- Integrated into POST /api/feedback with automated unit & integration test suites.

3. React Frontend & Visual Analytics (Phases 10 - 13)

- Modern React 18 + Vite + Tailwind CSS Single Page Application (SPA).
- Student Portal: Dashboard metrics, feedback submission with live NLP analysis modal, and history.
- Admin Dashboard: Interactive Recharts (Sentiment Pie, Category Bar, Department Stacked Bar, Trend Line), trending keyword cloud, actionable suggestions, and searchable feedback table.

4. Quality Assurance, Build & Viva Guide (Phases 14 - 16)

- NLP unit tests are included in tests/ml/; API integration tests are included in tests/backend/.
- Build locally with npm run build after installing frontend dependencies.
- Full documentation: system_design.md, data_dictionary.md, api_specification.md, and viva_preparation_guide.md.

CREDENTIALS & HOW TO RUN

FastAPI Backend

cd backend && uvicorn main:app --reload (Port 8000)

React Frontend

cd frontend && npm run dev (Port 5173)

Admin Account

Created on first startup from ADMIN_EMAIL and ADMIN_PASSWORD in backend/.env.

Student Account

Register a student account from the portal; public registration cannot create administrators.